

HEMACHANDRA REDDY POTTINGARI

pottingari@gmail.com | +91 63059 84164 | LinkedIn | GitHub (<https://github.com/Hemachandra9899>)

SUMMARY

AI and Full-Stack Engineer with extensive experience building production-grade AI-powered applications, LLM integrations, RAG pipelines, and scalable microservices using Python, TypeScript, FastAPI, Next.js, NestJS, and PostgreSQL. Experienced in vector databases (Pinecone, Qdrant), background job queues (BullMQ, Redis), containerization with Docker, and agentic workflows.

EDUCATION

Sreenidhi Institute of Science and Technology | 2021 – 2025
B.Tech. in Information Technology, Hyderabad, Telangana — CGPA: 7.14 / 10

TECHNICAL SKILLS

- Languages: Python, TypeScript, JavaScript, SQL, C, C++, Rust (Basics)
- Frontend: Next.js, React, Tailwind CSS, HTML5, CSS3
- Backend: FastAPI, NestJS, Node.js, Express.js, Fastify, REST APIs, Microservices
- AI and Data: LLM APIs (NVIDIA NIM, Groq, OpenAI), RAG Pipelines, Intent Detection, Prompt Engineering, Embeddings & Semantic Search, LangChain, Data Pipelines, Document Analysis
- Databases: PostgreSQL, Pinecone, Qdrant, MongoDB, MySQL, FAISS, Redis
- Tools & DevOps: Docker, Docker Compose, Git, GitHub Actions, CI/CD, BullMQ, Prisma

EXPERIENCE

AI and Data Engineer | AlignLabs | Feb 2026 – Present

- Built full-stack AI applications using Next.js, TypeScript, FastAPI, Python, and PostgreSQL, integrating LLMs and third-party APIs.
- Developed automated pipelines that collected marketing-platform data, transformed it, and populated structured Google Sheets for reporting and campaign tracking.
- Built an Ask AI system with intent detection, contextual retrieval, and answer generation for marketing and analytics queries.
- Developed analytics dashboards for campaign performance, platform metrics, and AI-generated business insights.
- Implemented AI-assisted workflows that performed actions such as generating content, updating records, creating reports, and sending automated emails.
- Built document-analysis features for information extraction, summarization, and question answering over uploaded enterprise documents.

SDET Intern | CodeNebula | Jul 2024 – Dec 2024

- Developed and tested web application features using React, Node.js, Tailwind CSS, HTML, and CSS.
- Performed end-to-end testing, API validation, debugging, and performance testing for an application serving 500+ active users.
- Created comprehensive test cases, documented defects, and worked with developers to verify fixes and prevent regressions.

KEY PROJECTS

Scout – Evidence-First AI Research Engine — <https://github.com/Hemachandra9899/Scout>

Tech: Next.js, TypeScript, FastAPI, PostgreSQL, Qdrant, Redis, BullMQ, Docker
• An evidence-first AI research engine that conducts intent routing, multi-source search, web crawling, evidence extraction, reranking, citation verification, and grounded answer generation.

Second Brain – AI Knowledge Management System — <https://github.com/Hemachandra9899/second-brain>

Tech: Next.js, NestJS, FastAPI, Pinecone, NVIDIA NIM, PostgreSQL, Redis, Docker
• A comprehensive personal knowledge-management platform featuring task management, semantic vector search, and AI-assisted information retrieval.

PDF RAG Assistant — <https://github.com/Hemachandra9899/Pdf-Rag-Assistant>

Tech: Python, LangChain, FastAPI, Vector DB, LLMs
• An intelligent document-analysis application that enables users to upload complex PDF documents and ask questions with contextual retrieval and grounded citations.

CogniTalk — <https://github.com/Hemachandra9899/CogniTalk>

Tech: Next.js, TypeScript, Tailwind CSS, LLM APIs
• A modern conversational AI application featuring streaming responses, custom personas, and responsive user interfaces.